Gears and wheels

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use gear ratio and wheel circumference to predict an ideal travel speed.

SUCCESS CRITERIA - I can...

1. Connect gear teeth with speed ratio.
2. Calculate distance per wheel revolution.
3. Explain the tradeoff between ideal speed and available torque.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gear ratio” and “circumference” to explain your model or result. Sentence starters are optional.

gear ratio: a comparison of gear tooth counts or rotation rates

circumference: distance around the edge of a circle

torque: the turning effect of a force about an axis

01 Warm-up: bring back an earlier idea

A motor uses 3 W for 5 s.
Find electrical energy and name a possible non-mechanical output.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

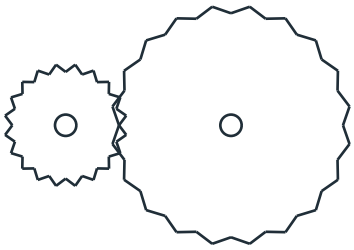
02 Worked example: explain the model

Reference: $rpm_{out} = rpm_{in} \times teeth_{driver}/teeth_{driven}$; $C = \pi d$
$v = C \times rpm_{wheel}/60$.
Ideal gears and rolling without slip; C in meters gives v in m/s. Loading and losses change real performance.

Worked reasoning: A 10-tooth driver turns a 30-tooth driven gear. The driven gear makes one turn for every three driver turns. In an ideal gear pair, slower rotation trades for greater torque.

Your turn: Draw both gears, label tooth counts and opposite rotation directions.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Driver Driven

Marzano check - circle: ① ② ③

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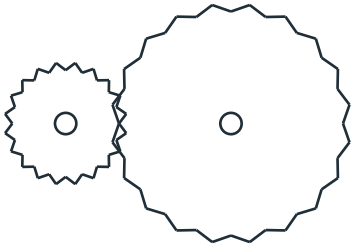
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A small driver gear turns a larger driven gear.
Predict which turns more slowly and indicate their opposite rotation directions.

Sentence starter: Both … . They differ in … because … .



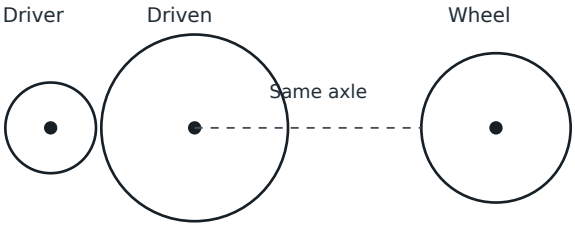
Marzano check - circle: ① ② ③

04 Apply the physics

A model car’s wheel circumference is 0.20 m and its wheel rate is 120 rpm.
Label the wheel model.
Convert rpm to revolutions per second, then calculate ideal ground speed with no slip.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Label teeth, rotation rates and wheel size.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A team chooses the largest wheels because the formula predicts more distance per turn.
Explain why measured acceleration or speed might still decrease.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A total current of 14 A splits between two identical parallel branches.
Find each branch current.
Draw the stated circuit and label quantities relevant to the question.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw the complete circuit and label shared nodes.

Marzano check - circle: ① ② ③

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Continue down each column.

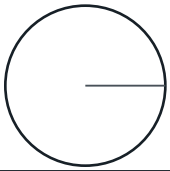
Sentence starters are optional. You may plan in any language.

07 Exit check

At the same wheel rpm, doubling circumference does what to ideal travel speed?
State one real-world limitation.

Word help: ideal = using stated simplifying conditions

Sentence starter: My claim is … . The diagram/data show …, so … .



Radius / diameter: ____

Turns per second: ____

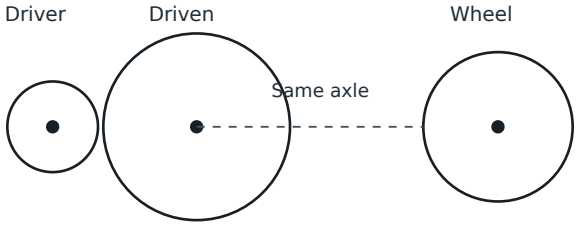
Travel per turn: ____ Ground speed: ____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Design two different tooth-count pairs giving the same 3:1 reduction.
Explain whether tooth counts alone determine the car's ground speed.

Sentence starter: Both … . They differ in … because … .



Label teeth, rotation rates and wheel size.

Marzano check - circle: 1 2 3